MIL-G-85742(AS)

4 FEBRUARY 1987

MILITARY SPECIFICATION

GUIDANCE CONTROL SECTION, GUIDED MISSILE

WGU-15(XCL-1)/B

(SIDEARM)

This specification is approved for use within the Naval Air System Command, Department of the Navy, and is available for use by all Departments and Agencies of the Deparment of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance for the Guidance Control Section, Guided Missile WGU-15(XCL-1)/B, critical item referred to herein as the "GCS".

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD) Form 1426) appearing at the end of this document or by letter.

FSC 1420

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1. Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. unLess otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-5	Capacitors, Fixed.
MIL-S-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapon Systems.
MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems.
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice, Process for).
MIL-S-7742	Screw Threads, Standard Optimum Selected Series; General Specification for.
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-S-8802	Sealing Compound, Temperature Resistant, Integral Fuel Tanks and Fuel Cell Cavities High adhesive.
MIL-M-8856	Missile, Guided, Strength and Rigidity, General Specification for.
MIL-T-11393	Test Set, Continuity, Rocket Igniter Circuit.
DOD-P-15232	Phosphate Coating, Heavy, Manganese of Zinc Base (for Ferrous Alloys).

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MIL-S-19500	Semiconductor Devices, General Specifications for.
MIL-P-23377	Primer Coating, Epoxy Polymide, Chemical and Solvent Resistant.
MIL-M-38510	Microcircuits, General Specification for.
MIL-C-45224	Cable and Harness Assemblies, Electrical, Missile System, General Specification for.
MIL-W-46132	Welding, Fusion, Electron Beam, Process for.
MIL-P-81001	Power supply PP-2581.
MIL-C-435285	Coating Polyurethane, Aliphatic Weather Resistant, Low Infrared (IR) Reflective.
MIL-G-85742/1	Seeker Section Assembly.
MIL-G-85742/2	Receiver Subassembly.
MIL-G-85742/3	Power Supply Assembly.
MIL-G-85742/4	G-Bias/Nav Gain Assembly.
MIL-G-85742/5	Servo Section.

STANDARDS

Federal

FED-STD-102	Preservation, Packaging, and Packing Levels.
FED-STD-595	Colors.

Military

MIL-STD-9	Screw Thread and Methods of Specifying.
MIL-STD-129	Marking for Shipment and Storage.

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MIL-STD-198	Capacitors, Selection and Use of.
MIL-STD-199	Resistors, Selection and Use of
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-701	List of Standard Semiconductor Devices.
MIL-STD-704	Electric Power, Aircraft Character and Utilization of.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests; Exponential Distribution.
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production.
MIL-STD-810	Environmental Test Methods.
MIL-STD-889	Dissimilar Metals.
MIL-STD-965	Parts Control Program.
DOD-STD-1399	Interface Standard for Shipboard Systems.
MIL-STD-1562	List of Standard Microcircuits.
MIL-STD-1595	Qualification of Aircraft, Missile and, Aerospace Fusion Welders.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-2175	Castings, Classification and Inspection Of.
MIL-STD-33450	Safety Wiring.
MIL-STD-33588	Nut, Self Locking, Aircraft Design and Usage Limitations of.
MIL-STD-45662	Calibration Systems Requirement.

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2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation

DRAWINGS

Naval Ordnance Systems Command
(Code Ident 10001)

1516568	Container, Shipping and Storage MK 270 MOD 1 (For TWO G & C Groups MK 12 MOD 2) General Arrangement.

Naval Air Systems Command
(Code Merit 30003)

1466AS1082	Seeker Section Assembly.
1466AS1200	Guidance Control Section WGU-15(XLC-1)/B.
146AS1202	Servo Section Assembly A3.
1466ASJ271	Interconnect Diagram GDNC Cont Sect.
1466AS1302	Power Supply Assembly.
1466AS1312	Receiver Subassembly.
1466AS1324	G-Bias/Nav Gain Assembly.

PUBLICATIONS

Naval Air System Command
(code Ident 30003)

AS-4613	Applicacation and Derating Requirements for Electronic Components.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

Naval Weapons Center
(Code Ident 12934)

NWC 1430 Appendix II	Electromagnetic Compatibility (EMC) Criteria for the Sidearm Missile.

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Code of Federal Regulations

49 CFR 171-178	Transportation.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for asso-elated specification sheets), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede appli-cable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Item description. The Guidance Control Section (GCS) provides the acquisition, tracking and guidance electrical and pneumatic/aerodynamic functions for the Sidearm Missile necessary to provide a proportional navi-gation course-to-target intercept. The GCS is comprised of a seeker assembly, a receiver assembly, a servo assembly, and m umbilical assembly. The GCS receives and guides on the radio frequency (RF) signals specified in the Appendix.

3.1.1 Government-furnished property. The Government will furnish the following property (see 6.5).

	Description	Nomenclature
a.	Guidance Control Group	MK-12 Mod 3
b.	rest set	MK-401 Mod 1
C.	Cages	
d.	Containers	MK 270 Mod 1

3.2 Characteristics.

3.2.1 Performance. The GCS system shall meet the function performance requirements specified herein.

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3.2.1.1 GCS operational currents. The following requirements are established for start-up and run of the GCS:

a. When 115 volts alternating current (Vac), 400 hertz (Hz) is applied to umbilical (connector P1004) pin 1 with return on pin 21, the missile current drain shall be less than 3.5 amperes root-mean-square (rms) during start-up. When 175 volts direct current (Vdc) is applied to umbilical pin 29 with return on pin 21, the current drain shall be less than 11 milliamperes (mA) for The 175 Vdc line.

b. Within 2 minutes, the run current shall be less than 1.75 amperes (A) maximum for the 115 Vac line and less than 11 mA for the 175 Vdc line.

3.2.1.1.1 Input voltage. The GCS shall be able to operate on voltages meeting the following requirements:

a. The 115 volts root-mean-square (Vrms), 400 Hz source as defined in MIL-STD-704 for captive carriage on aircraft and DOD-STD-1399, section 103 for ground power except the lower voltage limit shall be 104 Vrms.

b. The 175 Vdc source shall be as defiuned in MIL-P-81001 for the LAU 7/A launcher.

c. The 28 Vdc source shall be in accordance with MIL-STD-704.

3.2.1.2 Regulator output. The regulator output current at umbilical, connector P1004 pin 32 shall actuate the relay of Figure 1.

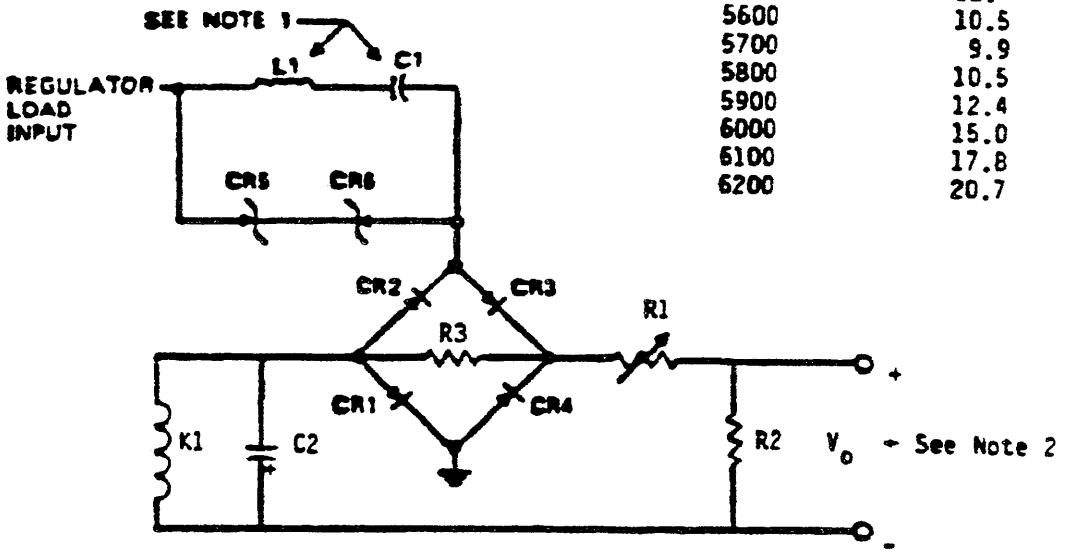
3.2.1.3 Launch lock Voltage. The GCS launch lock voltage on target detecting device (TDD) connector (A3A1W1P2) pin 7 (Drawing 1466AS1271) shall be greater than 12 Vdc when loaded with a load of 29.0 ±2.5 ohms.

3.2.1.4 Launch latch continuity. The resistance between common leads of the GCS TDD cable connector A3A1U1P2 in accordance with Drawing 1466ASL271 and the umbilical connector P1004 shall be not greater than 0.25 ohms.

3.2.1.5 Gas generator squib resistance. The squib circuit resistance shall be 3.25 ±0.50 ohms. Test current shall be less than 5 mA.

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FREQUENCY Hz	VOLTAGE RMS
5200	21.5
5300	18.1
5400	15.0
5500	12.4
5600	10.5
5700	9.9
5800	10.5
5900	12.4
6000	15.0
6100	17.8
6200	20.7



CIRCUIT COMPONENTS:

R1: 50 Ω ,ADJUSTABLE, 2W (SEE NOTE 2)
R2: 1 Ω ,1%, ½W
R3: 1.2kΩ ,5%, 2W
K1: RELAY; ALLIED MHX-4213 OR EQUIV. (59A44A320)
250 ± 25 Ω COIL RESISTANCE
C1: APPROXIMATELY 0.009 µF; 300 Vdc; per MIL-C-5.
C2: 22 µF; 100V
CR1,CR2,CR3,CR4: IN4004 OR EQUIV.; FORWARD CURRENT AT 1V, 400 mA; INVERSE CURRENT AT 300V, 0.2 µA; PIV-300 V
CR5,CR6,: ZENER DIODE, IN3014, 180V, 10W.
L1: TORODIAL CORE, A206D68-2; 1160 TURNS NOMINAL, NO. 32 WIRE; RESISTANCE, NOMINAL 16 Ω; INDUCTANCE, 78 to 82 mH (REMOVE TURNS TO OBTAIN INDUCTANCE VALUE.)

NOTES:

1. THE CIRCUIT SHALL BE TUNED TO 5700 ± 50 Hz with 15 VOLTS RMS OUTPUT BY SELECTION OF C1.

2. FOR AN OUTPUT VOLTAGE OF 25 mVdc THE INPUT VOLTAGE SHALL BE WITHIN ± 2 VOLTS OF THE VALUES GIVEN IN THE TABLE ABOVE, USING A SINE WAVE VOLTAGE GENERATOR. RESISTOR R1 MAY BE ADJUSTED AT THE FREQUENCY OF 5700 ± 15 Hz TO OBTAIN THE VALUES GIVEN IN THE TABLE.

FIGURE 1. Simulated interlock relay circuit.

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3.2.1.6 False seeker steering commands. While tracking a 2C power level, pulse repetition frequency (PRF) 2 and a pulse width (PW) of 2 target at 10 ± 2 degrees off boresight, the seeker torquer zero to peak and peak to peak (p-p) voltages at A3A1W1P1 pins 10 and 11 shall be not greater than 400 millivolts (mV) (1.33 degrees per second (deg/sec)) at freq~encies C through E, not greater than 600 mV (2.0 deg/sec) at frequencies B and F and not greater than 750 mV (2.5 deg/sec) at frequencies A + 0.5 gigahertz (GHz) and G - 0.5 GHz when the (XS is rolled clockwise and counterclockwise about boresight at an angular rate of 1 radian/sec. This requirement shall be met for 20 of the 28 torquer outputs measured at the 7 frequencies specified above. (See the Appendix for a definition of the radio frequency signal parameters).

3.2.1.7 Video output pulse amplitude. The video output pulse amplitude shall be 3.0 ±0.6 volts (V) for GCS input power levels from lower threshold +10 decibels (dB) to the power level at which the RF attenuator is enabled +3 dB. This requirements shall be met for signal conditions defined in the Appendix.

3.2.1.8 Target acquisition. The GCS shall provide the capability to scan the frequency band of frequency A to G and acquire a target with$n T1 + T2. The GES shall continue to re-scan until a target is acquired. After target acquisition and upon receipt of the uncage/recage command, the GCS shall be capable of rescanning and reacquiring the target.

3.2.1.9 Target reaquisition. Upon loss of target track the GCS shall start sweeping upward in frequency and acquire the next target to appear.

3.2.1.10 Field-of-view. The unambiguous field-of-view shall be not less than ±15 degrees (deg) from missile boresight.

3.2.1.11 Lock-on sensitivity and dynamic range. While aboard the aircraft the RF attenuator shall be enabled when the seeker is caged. While in this state the GCS shall have the capability to acquire and track targets over the frequency range of A to G whose power varies over at least the range from 2C to 1C. With the attenuator free to operate as directed by the AGC design (seeker uncaged) the GCS shall have the capability to acquire and track targets over the frequency range of A to G whose power level covers the range of at least 4B to 1D. Upon acquistion of targets over the power levels described, the GCS shall continue to track these targets to power levels of 1C + 10 dB or greater. The RF attenuator shall be enabled at 43 ± 5 dB above the lower threshold signal level, provide 20 ± 6 db of attenuation and provide 8 ± 6 dB of hysteresis.

3.2.1.12 Gyro steady-state speed and slowdown time. The gyro steady state speed with the gyro caged and the GCS stationary with the radome in place shall be as specified in the Appendix. Ten seconds after removing power to the gyro the speed shall decrease not less than 7 Hz and not greater than 20 Hz of the speed while powered.

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3.2.1.13 Loop gain/phasing.

3.2.1.13.1 Tracking loop gain. The tracking loop gain shall be between 2.5 deg/sec per degree of tracking error and 6 deg/sec per degree of tracking error for frequency A to frequency C, between 3 deg/sec per degree of tracking error and 6 deg/sec per degree of trucking error for frequency C to frequency E and between 3 deg/sec per degree of tracking error and 10 deg/sec per degree of tracking error for frequency E to frequency G. The requlremente shall be met for 10 of the 14 measurements at power level 2C.

3.2.1.13.2 Tracking loop phasing. GCS phasing shall be adjusted such that the out of plane tracking error is less than 10 percent of the in plane tracking error.

3.2.1.13.3 Caging loop gain. The caging shall be 8 ±2 deg/sec per degree of gyro lag.

3.2.1.14 Pilot tone. When system threshold has been exceeded, the GCS shall generate the pilot's tone which shall be the PRF of the incoming signal. The pilot tone signal shall be a pulse 100 ±15 microseconds wide, 3.8 ±0.2 V p-p at the PRF of the incoming signal. With no signal present the number of false threshold crossings shall be less than 5 in a 10 second interval.

3.2.1.15 Servo torque output. The servo torque output shall be greater than 1,050 inch-pounds (in-lb) while tracking a target moving at a rate of 6 deg/sec.

3.2.1.16 Servo torque balance. The servo imbalance measured in the up-down and in the right-left channels shall be not greater than 50 in-lb.

3.2.1.17 Servo output noise. While tracking a stationary target positioned on an extension of longitudinal axis of the GCS, which is operating on internal per, no more than five servo torque peaks in excess of 100 in-lb in any one direction, as measured from measured from the average levels, shall be observed during any 2 second interval.

3.2.1.18 System gain. The system gain in each of the four quadrants) shall be 270 ±70 in-lb servo torque output while tracking a target moving at a planar rate of 1 deg/sec.

3.2.1.19 Average system gain. The average system gain for each channel shall be 270 ±50 in-lb in servo output torque per deg/sec planar tracking rate.

3.2.1.20 Steady-state cross talk. The p-p value of cross talk in the transverse plane shall be not greater than 200 in-lb while tracking a target moving at a planar rate of 6 deg/sec in the opposite plane,

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3.2.1.21 Servo disabled-enabled operation. During operation on internal power and for at least 0.4 ± 0.1, -0.0 second after removal of disabling voltage, the servo output torque in each channel shall be not greater than 50 in-lb. Servo torque shall reach 65 percent of the steady-state value between 0.4 and 0.8 sec after removal of the magnetic amplifier disabling voltage.

3.2.1.22 Enabling transient. The amplitude of the enabling transient torque shall be not greater than 50 in-lb in each channel.

3.2.1.23. G-bias/navigation gain. With the GCS caged, in any of the three positions of Figure 2, and servo magnetic amplifier enabled, the G-bias/navigation gain function torque in each fin plane shall be as shown in Figure 3 upon removal of the G-bias inhibit voltage and the navigation set voltage.

3.2.1.23.1 G-bias torque vector. The G-bias torque vector, as established in the roll phasing (see Figure 2), shall be within ±11 degrees of the up direction at 10 ±1 seconds after removal of the G-bias inhibit voltage.

G-bias vector angle - 45 - arc tan (torque right fin/torque up fin).

3.2.1.24 Power changeover and umbilical disconnect. Power changeover and umbilical disconnect shall not affect seeker target lock-on and tracking.

3.2.1.25 AFC signal. With no RF signal and internal power applied and the magnetic amplifier enabled, the automatic frequency Control (AFC) voltage shall be continous with no breaks.

3.2.2 Physical characteristics Physical characteristics of the GCS shall be as specified on drawing 1466AS1200.

3.2.3 Reliability.

3.2.3.1 Reliability program. A reliability program shall be established in accordance with MIL-STD-785. Unless otherwise specified, the reliability program plan shall encompass, as a minimum, tasks 101, 102, 103, 104, 105, 202, 203, 207, 208, 301 and 303 of MIL-STD-785 (see 6.2.2).

3.2.3.2 Quantitative reliability requirements. The GCS shall be capable of meeting a Mean-Time-Between-Failure (MTBF) Θ_0 of 200 hours at the 90 percent lower confidence limit. Any Government furnished material used in the GCS shall be excluded from the MTBF requirements.

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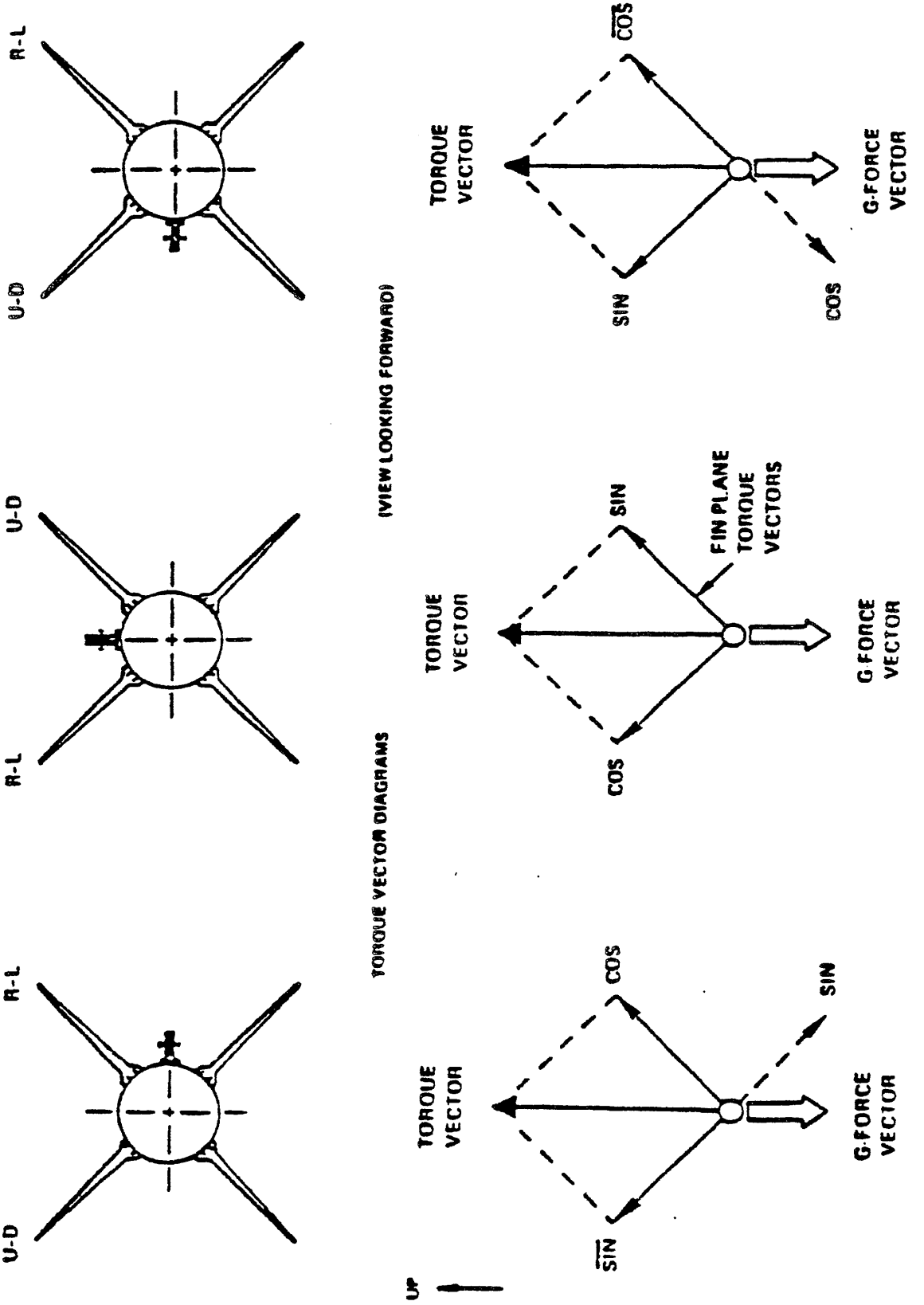


FIGURE 2. G-bias roll phasing.

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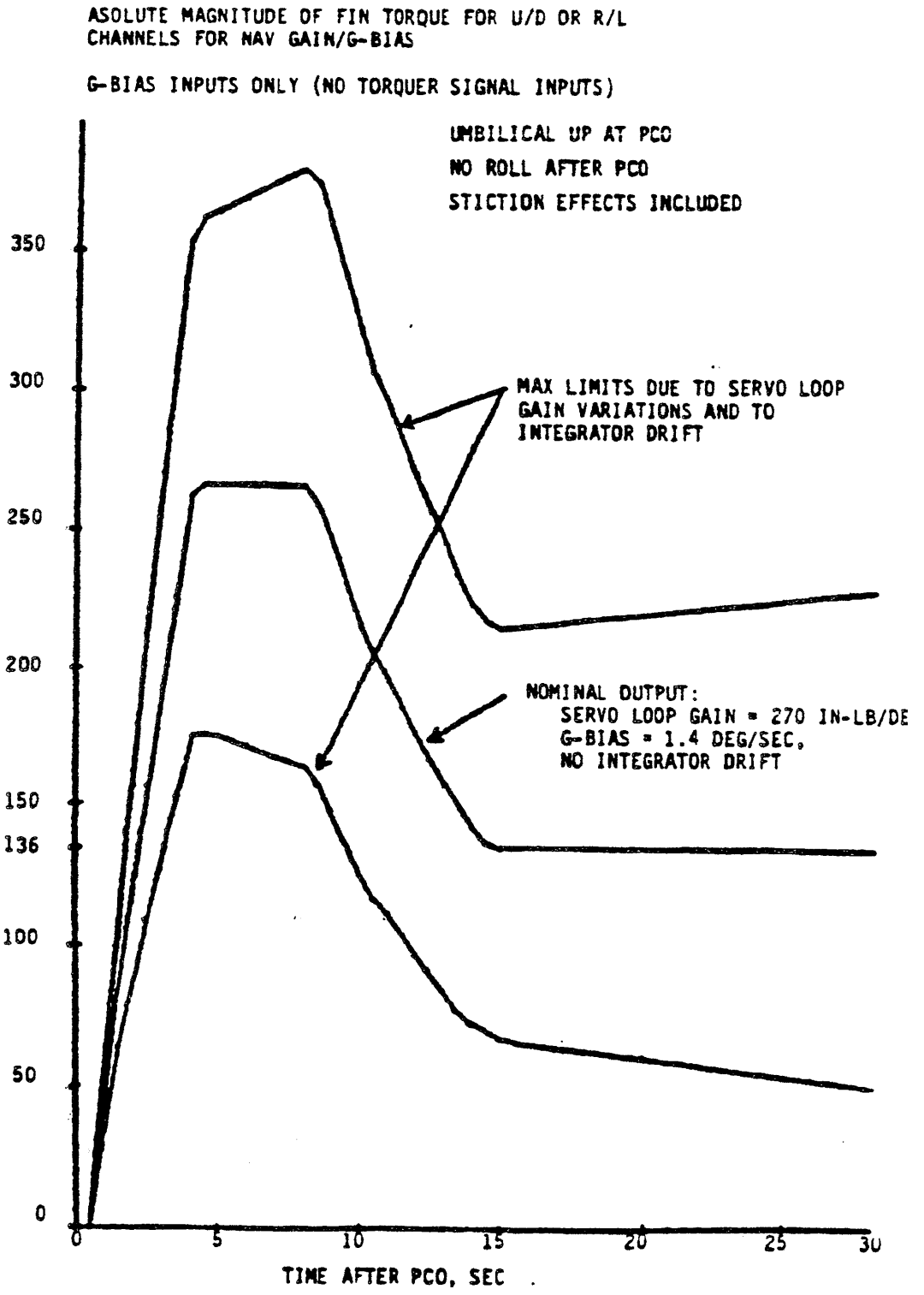


FIGURE 3. Variable nav gain/G-bias function.

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3.2.3.3 Reliability test requirements. Prior to release for limited or unlimited production, the GCS shall demonstrate the capability of meeting the required MTBF (Θ_0), as specified in 3.2.3.2.

3.2.3.3.1 Reliability demonstration (MTBF). GCS assemblies supplied in accordance with this specification shall be capable of demonscrating a minimum MTBF as specified in 3.2.3.2 when tested in accordance with MIL-STD-781, Test Plan XIVC, unless otherwise specified in the contract or purchase order (see 6.2.1).

3.2.3.3.1.1 MTBF performance requirements. The MTBF performance requirements shall meet the requirements of 3.2.1 when subjected to the tests specified herein.

3.2.3.3.1.2 UTBF determination. The GCS operating time shall be used An MTBF determinations. Each GCS operating time shall be not less than one-half of the average operating time of aLL GCS which were utilized In the MTBF demonstration. For the purpose of MTBF determination, the GCS operating the shall be that time during the thermal cycle when the (GCS is energized.

3.2.3.4 Environmental stress screening. Each GCS shall be subjected to 100 percent screening as a part of the manufacturing process in order to disclose workmanship defects. The screening process shall consist of exposure to thermal cycling and random vibration.

3.2.4 Environmental. The GCS shall show no evidence of physical damage and shall meet the performance requirements of 3.2.1 before and after exposure to the following environmental conditions.

3.2.4.1 High temperature (nonoperating and operating). The GCS shall withstand storage at 71 degree Celsius (60°C) and must be operational at 60°C.

3.2.4.2 Low temperature (operating). The GCS shall withstand storage and be operational at -53°C.

3.2.4.3 Temperature shock (nonoperating). The GCS shall withstand sudden temperature transitions from 71 to -40°C and from -40 to 71°C.

3.2.4.4 Flight acceleration (operating). The GCS shall be operational during a longitudinal acceleration of 25 g and 28 g lateral axis acceleration.

3.2.4.5 Transportation vibration (nonoperating). The GSC shall be operable following exposure to 3 hours of vibration as shown in Figure 4.

3.2.4.6 Captive flight vibration (maneuver). The GCS shall be operable during and after exposure to vibration as shown in Figure 5.

3.2.4.7 Captive flight vibration (high speed dash). The GCS shall be operable during and after exposure to vibration as shown in Figure 6.

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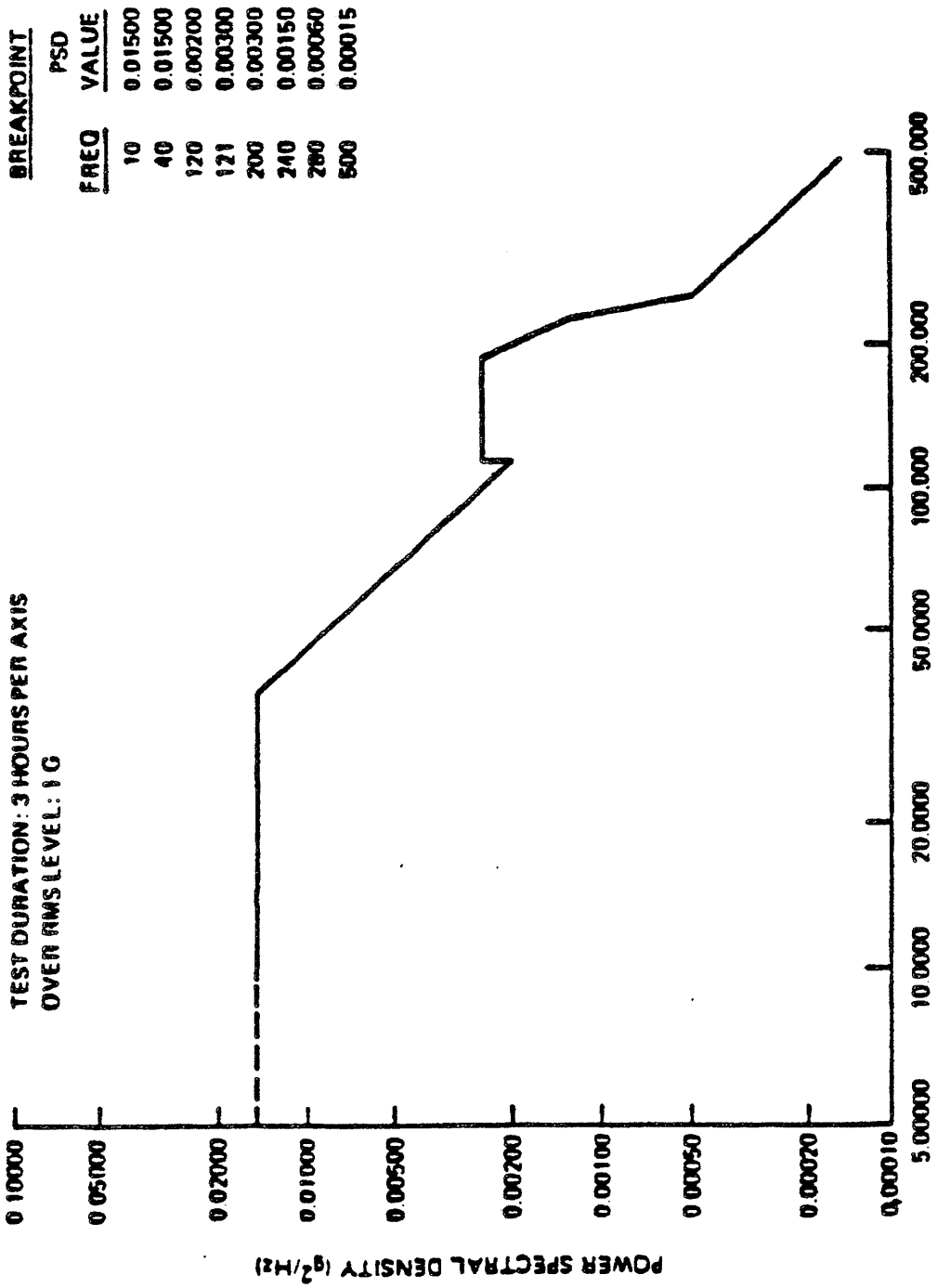


FIGURE 4. Basic transportation, common carrier environment.

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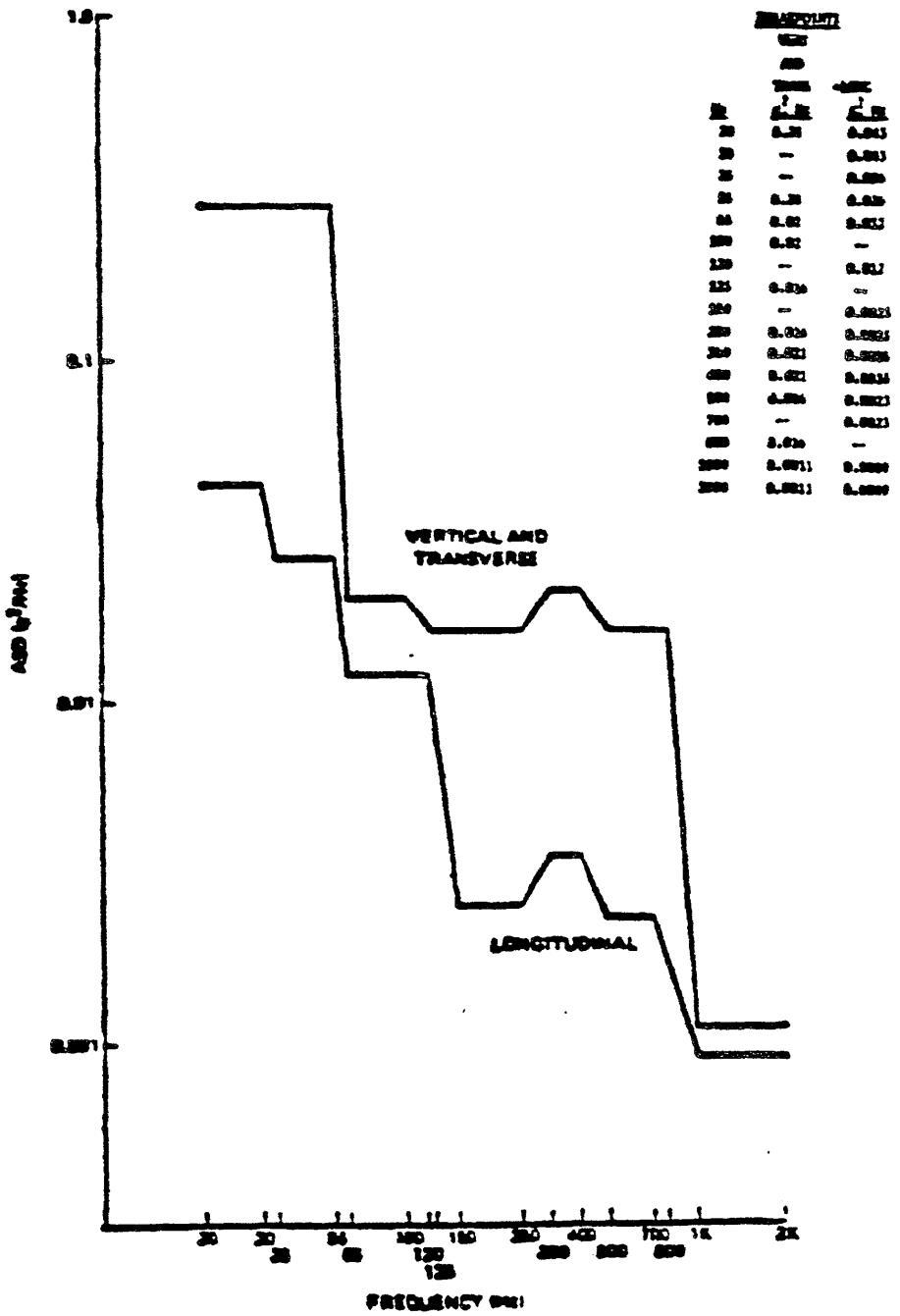


FIGURE 5. Captive flight vibration test requirement fixed wing aircraft, maneuver.

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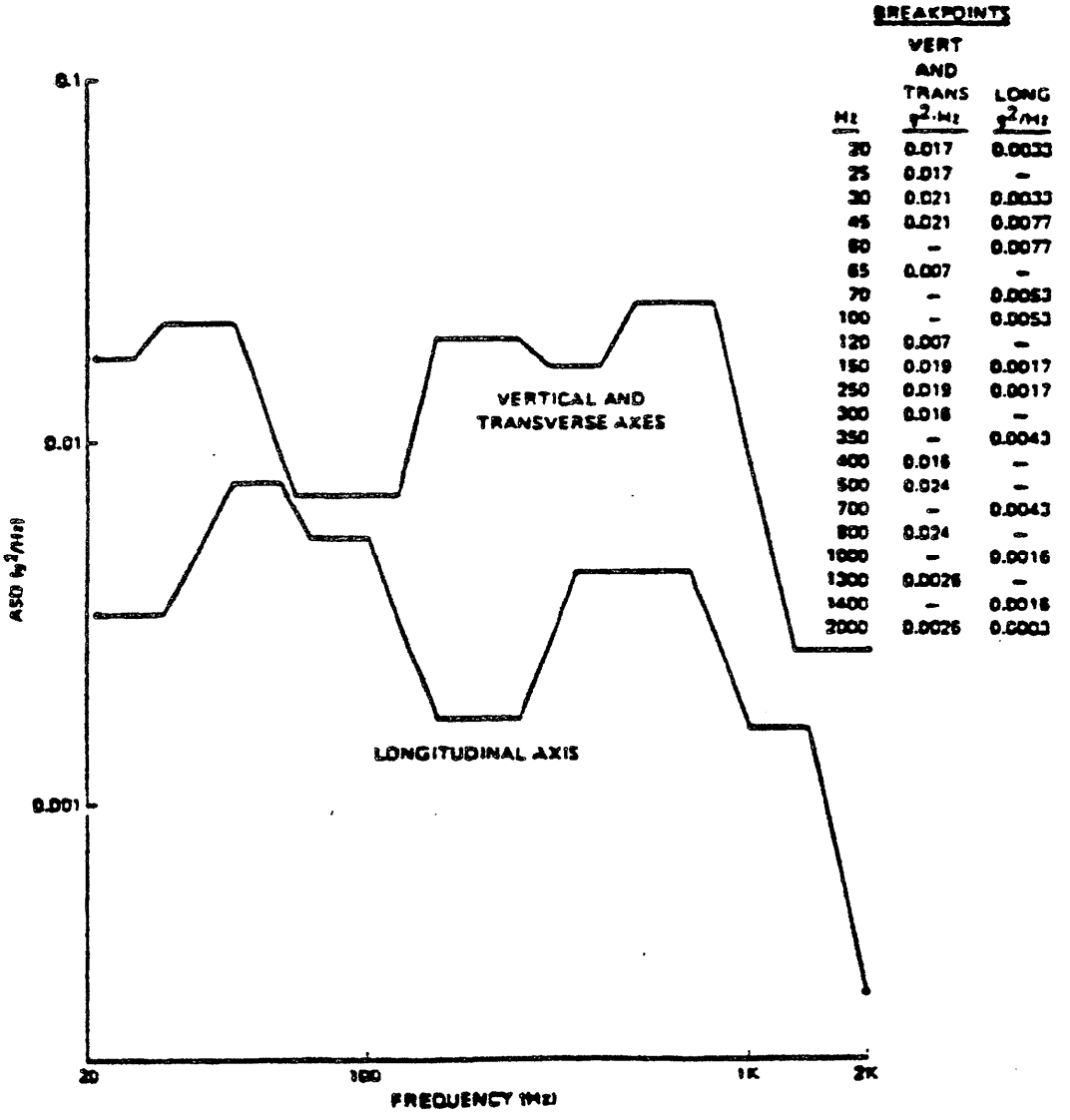


FIGURE 6. Captive flight vibration test requirement fixed wing aircraft, high-speed dash.

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3.2.4.8 Captive carry vibration helicopter (operating). The GCS shall be operable during and after exposure to sinusoidal vibration specified in Table I.

TABLE I. AH-1 helicopter captive vibration requirements.

Axis	Peak g			
	11 Hz	22 Hz	33 Hz	44 Hz
Vertical	2.9	2.7	1.3	1.8
Horizontal	1.1	1.9	1.6	1.0
Longitudinal	1.2	1.1	1.3	0.8

3.2.4.9 Transit shock (operating). The GCS, in the shipping container, shall withstand shock induced by dropping 48 inches onto a hard surface.

3.2.4.10 Ignition or tip off shock. The GCS shall be operable during and after exposure to longitudinal axis shocks of 30 g by 8 ms half sine pulse, and lateral axis terminal sawtooth pulses of 35 g amplitude by 15 ms duration.

3.3 Design and construction. All components comprising the GCS shall be of a minimum practical size and weight consistent with reliable operation. Construction, strength, and rigidity shall be as specified in MIL-H-8856.

3.3.1 Production drawings. The GCS shall be fabricated and assembled in accordance with the drawings, parts list, and ocher documents listed on Drawing 1466AS1200.

3.3.2 Standards of manufacture. Materials and parts shall be selected for their lightweight construction and simplicity of manufacture commensurate with the design of the item. Unless necessary for the design application, critical and strategic materials shall not be used. No cleaning or plating processes shall be used which may degrade the structural integrity of the GCS due to hydrogen embrittlement or stress corrosion. Materials not anticipated to be commercially available during the planned production period shall not be used. No substitution is authorized without prior approval of the procuring activity. Proprietary items or processes shall not be used.

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3.3.2.1 Processes.

3.3.2.1.1 Protective coatings and surface treatments. Unless otherwise specified herein, finishes of machine, cast, forged, and welded surfaces shall be the minimum required for proper operation, good appearance, and economy of manufacture. Protective coatings shall not be used in areao which will lose that coating during the normal course of operation and testing. Protective coatings shall be selected taking into consideration the environmental conditions imposed herein. Screws, bolts, slip-fits, screw-threaded inserts and press-fits shall be assembled using wet epoxy primer complying with MIL-P-23377 or wet sealing compound complying with MIL-S-8802.

3.3.2.1.2 Finishes. Finishes of metal surfaces shall be as specified in the applicable drawings. Coatings for exterior surfaces shall comply with the requirements of MIL-C-85285. The exterior finish of the GCS shall be green paint conforming to color 34095 specified in FED-STD-595.

3.3.2.1.3 Plating. Unless otherwise specified herein, plating of metal surfaces shall be in accordance with the requirements of MIL-S-5002. Zinc plating shall not be used due to its low resistance to corrosion.

3.3.2.1.4 Phosphate coatings. Phosphate coating for unpainted ferrous metal surfaces shall comply with the requirements of DOD-P-16232.

3.3.2.1.5 Dissimilar metals. Dissimilar metals shall not be used in direct concact unless suitably protected to prevent electrolytic corrosion. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-889. The selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-454, Requirement 16.

3.3.2.1.6 Brazing and welding. Brazing of the metals shall comply with the requirements of MIL-B-7883. Welding and welding processes shall comply wich the requiraments of MIL-W-46132 or MIL-W-8611. Welding operators shall be qualified in accordance with MIL-STD-1595.

3.3.2.1.7 Passivation. Unless otherwise specified herein, corrosion and heat-resistent be processed and passivated in accordance with the requirements of MIL-S-5002.

3.3.2.1.8 Heat treatment. Unless otherwise specified herein, heat treatment shall comply with the requirements of MIL-H-6875.

3.3.2.2 Parts.

3.3.2.2.1 Threaded parts. Threading of parts shall conform to the requirements of MIL-S-7742 and MIL-STD-9. Duplicate parts, differing only in thread form, shall not be permitted. When an allowance is required for application in elevated temperatures, corrosive atmosphere, or other conditions which may cause thread seizure, this allowance shall be obtained by increasing the diameter of the internal threads.

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3.3.2.2.2 Locking provisions. All threaded parts such as nuts, screws, and bolts shall be provided with safety wiring or other acceptable means of locking. Any adjustments which may be provided shall be capable of being locked securely enough to prevent disturbance of setting under all service and test conditions. Self-locking nuts shall conform to MIL-N-25027 and shall be used in accordance with MIL-STD-33588. Safety wiring shall comply with the requirement of MIL-STD-33450. The use of staking and lock washers shall be avoided unless they can be shown to have characteristics superior to current conventional methods and approval is obtained from the procuring activity.

3.3.2.2.3 Castings. Castings, if used, shall comply with MIL-STD-2175, Class 1, Grade 1.

3.3.2.2.4 Standard parts. The part classifications specified in 3.3.2.2.5 shall be considered as being standard parts for the electronics:

3.3.2.2.5 Standard parts. Standard parts as classified herein shall be used to the fullest extent possible. Parts falling outside of this classification are considered to be nonstandard; their use shall require justification, specification, and approval by the procuring activity.

a. Semiconductor devices. All semiconductor devices, including, but not limited to, transistors and diodes shall be selected from MIL-STD-701, qualified and procured in accordance with MIL-S-19500 as JANTX devices to the maximum extent possible.

b. Microelectronics. All microelectronic devises shall be selected from MIL-STD-1562 qualified and procured in accordance with MIL-M-38510 as Class B devices and shall be selected from a current Qualified Products List (QPL).

c. Passive piece parts. All piece parts, including, but not limited to, capacitors, resistors, connectors and relays shall be procured as Established Reliability (ER) types with an R failure rage level or better. All resistors shall be selected from MIL-STD-199 and the QPL associated with the selected resistor style. All capacitors shall be selected from MIL-STD-198 and QPL associated with the selected capacitor style.

3.3.2.2.6 Nonstandard parts. All parts not selected in accordance with 3.3.2.2.5 shall be considered as being nonstandard for the GCS. The use of nonstandard parts shall require approval by the procuring activity in accordance with MIL-STD-965, Procedure I. Nonstandard MIL parts and standard commercial parts shall require screening.

3.3.2.2.7 Connectors. Electrical connectors shall be selected and used in accordance with MIL-STD-454, Requirement 10.

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3.3.2.2.8 Application and derating of electronic components. The application and derating and electronic components used in the assembly shall be accomplished in accordance with AS-4613 requirements for Class B equipment. On any design changes to product baseline wherein present components are being changed or new components are being added parts shall be selected in accordance with the derating criteria of AS-4613 for Class A equipment.

3.3.2.3 Wiring harness and connectors. The wiring harnesses and connectors for the assembly internal and external interface connections shall be in accordance with MIL-C-45224. Bonding shall be in accordance with MIL-B-5087.

3.3.3 Soldering. Procedures and requirements for preparation and soldering of electrical connections shall meet the requiremencs of US-6536 unless otherwise specified in the contract or purchase order (see 6.2.1).

3.3.4 Special working environment. The contractor shall provide adequate facilities for the fabrication, assembly, and testing of supplies to be delivered in accordance with this specification which meet the requirmerits of MIL-STD-1695.

3.4 First article sample. When specified, a sample shall be subjected to first article inspection (see 4.1.2.1 and 6.3).

3.5 Electromagnetic compatibility (EMC). An overall integrated EMC Program, in accordance with WC 1430 Appendix H, for the system shall be established as specified in the contract or purchase order. (See 6.2.2)

3.6 Workmanship. Workmanship, fabricating, and assembly practices shall meet the requirements of MIL-STD-454, Requirement 9. The GCS, including all parts and accessories shall be constructed and finished in a manner go assume compliance with all requirments of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring. impregnation of coils, marking of parts and assemblies, plating, rivetings, machine screw assemblies, and freedom from burrs and sharp edges. All parts, components and the GCS shall be thoroughly clean of loose, spattered or excess solder, rosin flash, metal chips, and foreign matter.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right ot perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and service conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absense of my inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Special tests and examination. All GCS's to be used in special tests and examinations shall be subjected first to inspections and zests specified in Table II, Groups A, B, and C, to verify conformance to this specification. Disposition of the samples tested shall be as specified in the contract or purchase order (see 6.2.1).

4.1.2.1 First article inspections. When specified in the contract or purchase order, first article inspections shall be performed by the contractor after award of contract and prior to production, at a location acceptable to the Government (see 6.3). First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.1.2.2 Physical configuration audit. A Physical Configuration Audit (PCA) will be conduted on two GCS'S selected by the Government at random. The PCA will be performed following the procedures outlined in MIL-sTD-1521. (See 6.2.2)

4.1.2.3 Reliability demonstration test. The reliability demonstration test shall be conducted utilizing not less than four GCS's, unless otherwise specified in the contract or purchase order (see 6.2.1). Before and after exposure to the environmental requirements of figure 9, each test sample shall meet the requirements of 3.2.1. During power on, voltages, current and gyro speeds shall be monitored (see 6.2.2). The MTBF demonstration test shall be conducted in accordance with MIL-STD-781, Test Plan XIV C and the following:

a. Environmental exposure and test profile shall be in accordance with Figure 7.

b. GCS performance measurements:

(1) During thermal cycling, the following parameters shall be continuously monitored:

(a) Input current (see 3.2.1.1).

(b) Applied voltage (see 3.2.1.1.1).

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TABLE II. Quality conformance inspections.

Examination or test	Requirement paragraph	Method paragraph
Group A		
Visual inspection	3.6	4.3.1
Group B		
Electrical tests		
GCS operational current	3.2.1.1	4.3.2.1
Regulator output	3.2.1.2	4.3.2.2
Launch lock voltage	3.2.1.3	4.3.2.3
Launch Latch continuity	3.2.1.4	4.3.2.4
Gas generator squib resistance	3.2.1.5	4.3.2.5
False seeker steering commands	3.2.1.6	4.3.3.1
Video output pulse samplitude	3.2.1.7	4.3.3.2
Target acquisition	3.2.1.8	4.3.3.3
Target reacquisition	3.2.1.9	4.3.3.4
Field-of-view	3.2.1.10	4.3.3.5
Lock-on sensitivity and dynamic range	3.2.1.11	4.3.3.6
Gyro steady-state speed and slowdown time	3.2.1.12	4.3.3.7
Tracking loop gain	3.2.1.13	4.3.3.8
Pilot tone	3.2.1.14	4.3.3.9
Servo torque output	3.2.1.15	4.3.3.10.1
Servo torque balance	3.2.1.16	4.3.3.10.2
Servo output noise	3.2.1.17	4.3.3.10.3
System gain	3.2.1.18	4.3.3.11
Average system gain	3.2.1.19	4.3.3.12
Steady-state cross-talk	3.2.1.20	4.3.3.13
Servo disabled-enabled operation	3.2.1.21	4.3.3.14
Enabling transient	3.2.1.22	4.3.3.15
G-bias test	3.2.1.23	4.3.3.16
Power changeover and umbilical disconnect	3.2.1.24	4.3.3.17
AFC signal	3.2.1.25	4.3.3.18
Group C		
Environmental screening		
Temperature cycling	3.2.3.4	4.4.1
Random vibration screening	3.2.3.4	4.4.2

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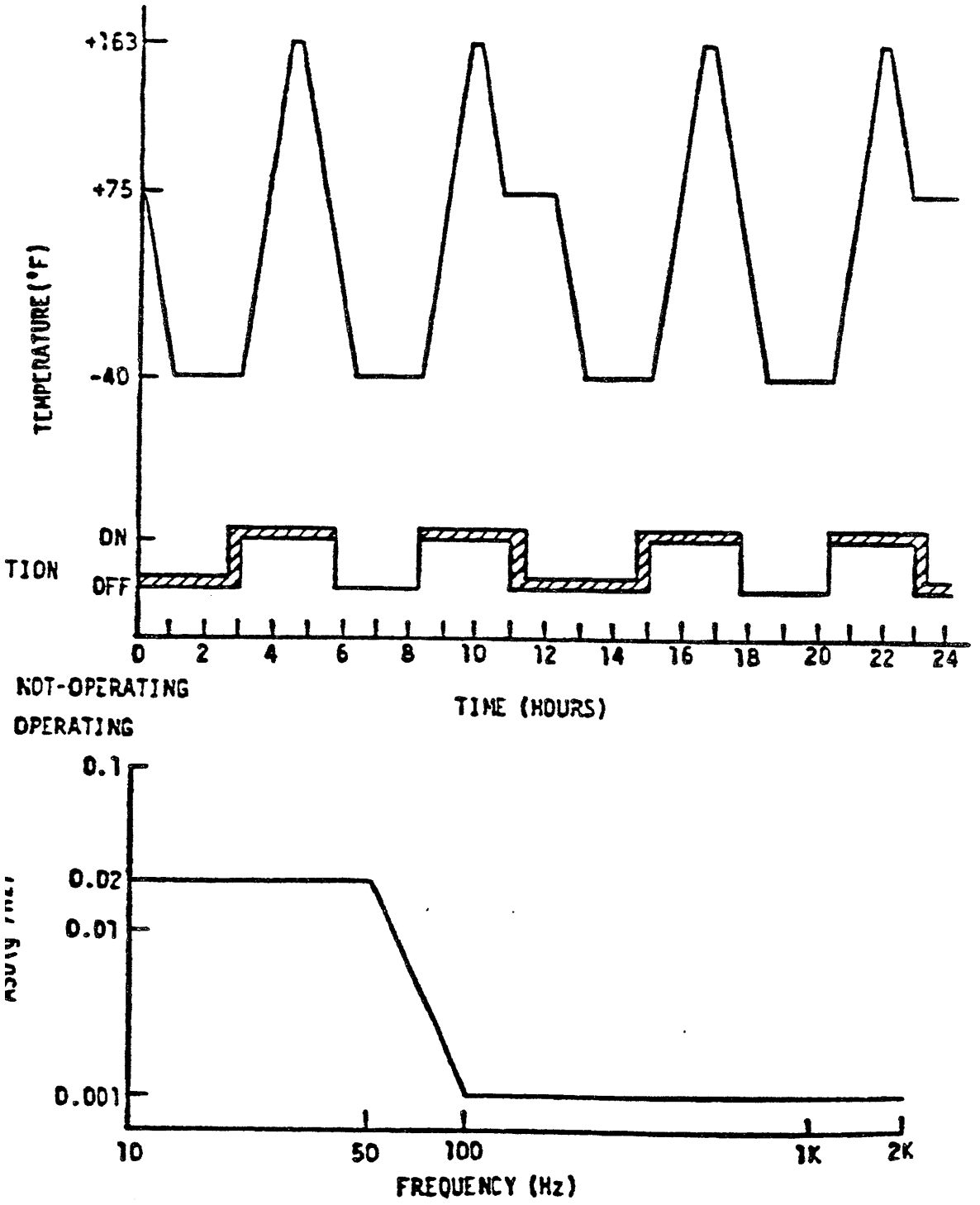


FIGURE 7. MTBF.

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(c) Gyro steady state speed (see 3.2.1.12).

(d) APC signal (see 3.2.1.25).

(e) Pilot tone (see 3.2.1.14).

(2) A performance test shall be perfomed at the conclusion of 2 ± 1 test cycles of 24-hours as shown on Figure 9. The GCS shall conform to the requirements and limits specified in 3.2.1.

d. Eighteen hours of GCS operating time shall be accumulated during each 24-hour cycle shown on Figure 9. In the event a failure is detected during chamber testing, all operating time accrued to the point of failure will be counted. a failures detected during performance verification. All operating time and all relevant failures of all GCS's committed to MTBF testing shall be used in MTBF determination.

4.1.3.2.2 Environmental qualification Unless otherewise specified the contract or purchase order (see 6.2.1), four GCS assemblies shall be selected from first article production units and subjected to the testing in accordance with contract requirements and 4.5

4.2 Quality conformance inspections and tests. All GCS's offered for acceptance shall be subjected to the quality conformance inspections and tests listed in Table I, Groups A, B, and C.

4.2.1 Test equipment calibration. All electrical, electronic, and mechanical measuring equipment used in acceptance testing and mechanical assembly shall be calibrated in accordance with a system meeting the requirements of MIL-STD-456624 Certification to this effect shall accompany all reports of first article, quality conformance testing, and reliability demonstration testing. Recalibration shall be made at necessary intervals or as required by the Government representative.

4.2.1.1 Test equipment and facilities. The contractor shall furnish and maintain all necessary test equipment, facilities, and personnel for the performance of all contractor-performed tests. When definite requirements are not specified, the test equipment shall be of sufficient accuracy and quality to permit satisfactory performance of the required acceptance tests.

4.2.1.2 Test equipment accuracies. Unless specific exceptions are made by the procuring activity or unless otherwise specified herein, test equipment calibration procedures shall be of such a nature and design that any errors produced to not greater than 15 percent of the maximum allowable tolerance range when both upper and lower limits are given, or 10 percent of the limit when only one limit (either upper or lower) is given.

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4.2.1.2. Tracking rate equipment. Test equipment used to induce tracking rates as specified herein shall have a tracking rate accuracy of ±5 percent.

4.2.1.2.2 Servo torque measuring equipment. The servo torque measuring equipment shall be of such a nature and design that the accuracy of each reading is within ±3 percent of the requirement specific range or 5 in-lb, whichever is greater. The measurement system frequency response shall be ±1 db from direct current (dc) to 25 Hz with the decreased response at 50 Hz not more than 3 db.

4.2.1.2.3 Voltage and current measurements. Unless otherwise specified herein, rms and dc voltage and current measurment inaccuracies shall be not greater than the lesser of 3.0 percent of the reading for alternating current (ac), 1.2 percent of reading for dc, or that required to meet the total measurement accuracy of 4.2.1.2. The low frequency cutoff (3 dB) down) of the rms voltmeter shall be less than 15 Hz and greater than 1 Hz. The high frequency cutoff (3 dB down) shall be greater than or equal to 100 kHz. Unless otherwise specified herein, equipment used to measure rms values shall be true rms responding.

4.2.1.3 Special test equipment. Unless specific exceptions are made by the procuring activity or unless values are given for individual tests, all special test equipment employed as a means for testing the GCS shall be of such a nature and design that any errors produced do not exceed 10 percent of the maximum allowable tolerance established for the value to be measured.

4.2.1.4 Test range. A test range shall be provided such that the seeker shall be in the far field of the radiating source (target). Spurious reflections in a 2-foot diameter quiet zone centered on the gimbal axes shall be at least 30 dB below the direct illumination.

4.2.2 Test environment. Unless otherwise specified in the applicable Government approved test procedures, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): 23 ±5 degrees Celsius (°C).

b. Altitude: Elevation of test facility.

c. Relative humidity: 95 percent or less.

4.2.3 GCS operating conditions. Unless otherwise specified herein, operation of the GCS shall be in accordance with the conditions specified in 4.2.3.1.

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4.2.3.1 Power. When required, GCS operating power shall be in accordance with the following:

a. External operational power. External power shall be applied by external electrical power supplies meeting the following requirements:

(1) 400 Hz power: 115 -11, +5 Vrms, 400 ±20 Hz single phase electrical power applied to pin 1 of P1004 (with ground return at pin 21).

(2) 28 Vdc: 28 +1, -6 V applied to pin 18 of P1004.

b. External control voltage:

(1) Magamp control voltage: +175 ±7 Vdc applied to pin 29 of P1004 (umbilical connector) (with ground return at pin 21).

c. Internal power. Internal power shall be supplied by a high-pressure air or nitrogen gas supply meeting the following requirements:

(1) Servo air: Gas pressure shall be 1150 ±50 psi at the GCS with GCS servo enabled. The gas shall have a dew point not greater than -35°C. The maximum particle size in the servo air shall be not greater than 30 microns. A flow rate of approximately 30 cubic feet per minute at standard temperature and pressure is required. High-pressure air or nitrogen is required only for tests involving torque measurements and tests involving internal power.

(2) Internal, power conditions: An internal GCS power condition (power changeover) will be established when the relay shown on Figure 1 is enabled removing the external power.

4.2.3.2 Mechanical load. During all tests where servoair is applied, each GCS rocker arm shall be mechanically loaded by an undamped spring having a constant of 80 ±5 in-lb per degree of rocker arm rotation. Restraint may be applied to either or both ends of the rocker arm

4.3 Inspection and test methods.

4.3.1 Visual inspection. The GCS assembly shall be visually inspected to verify conformance to 3.2.3, 3.3 and subparagraphs thereof, and the workmanship requirements of 3.6. Any CCS, or subassembly thereof, that does not meet the foregoing requirements shall be rejected.

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4.3.2 Electrical tests.

4.3.2.1 GCS operational currents. Measure and record the following current values:

a. 115 Vac start-up: The 115 Vac start-up current (pin 1 P1004) shall conform to 3.2.1.1(a).

b. 175 Vdc start-up: The 175 Vdc start-up current (pin 29 P1004) shall conform to 3.2.1.1(a).

c. Run current. The run current shall conform to 3.2.1.1(b).

4.3.2.2 Regulator output. With magamp enabled, by removal of the +175 Vdc, apply internal power in accordance with 4.3.2. Load umbilical P1004 pin 32 with the load shown on Figure 1 and use umbilical P1004 pin 21 as ground return. Output shall meet the requirments of 3.2.1.2.

4.3.2.3 Launch lock voltage. Connect an external 29.5 ± 2.5 ohms load from launch lock to ground. With the magamp enabled, operate the GCS on internal power. The launch lock voltage shall be not less than 12 Vdc across a 29.5 ± 2.5 ohms resistive load. The ac ripple shall be less than 2 volts p-p (see 3.2.1.3).

4.3.2.4 Launch latch continuity. Measure the resistance across pin 6 of connector A3A1W1P2 and pin 3 of umbilical connector P1004. Resistance shall meet the requirements of 3.2.1.4.

4.3.2.5 Gas generator squib resistance. Measure the gas grain squib resistance between pin 13 and pin 9 of the umbilical connector to determine conformance to 3.2.1.5 using a test circuit that meets the current requirements of MIL-T-11393.

4.3.3 Performance tests. Unless otherwise specified herein, the GCS shall be illuminated by a target (RF source) with the following characteristics: Frequency D, PRF 2, Power Level 2C as defined in the Appendix. The GCS shall be held with the longitudinal axis in a horizontal position (as shown on Figure 8) and mounted in a holding fixture subject to approval by the procuring activity.

4.3.3.1 False seeker steering commands. Mount the GCS in a fixture arranged to rotate the GCS about its longitudinal axis in a clockwise direction at a rate of 1.00 ± 0.10 radians/sec. With the signal conditions of 4.3.3 applied, the GCS shall track a stationary target positioned 10 ±2 degrees off boresight. While tracking the target rotate the GCS through at least four revolutions recording the steering voltages developed at A3A1W1P1 pins 10 and 11. Repeat this test for frequencies of A +0.5 GHz, B, C, E, F, and G - 0.5 GHz. Repeat this test while rotating in 8 Counter-clockwise direction. The results shall meet the requirements of 3.2.1.5.

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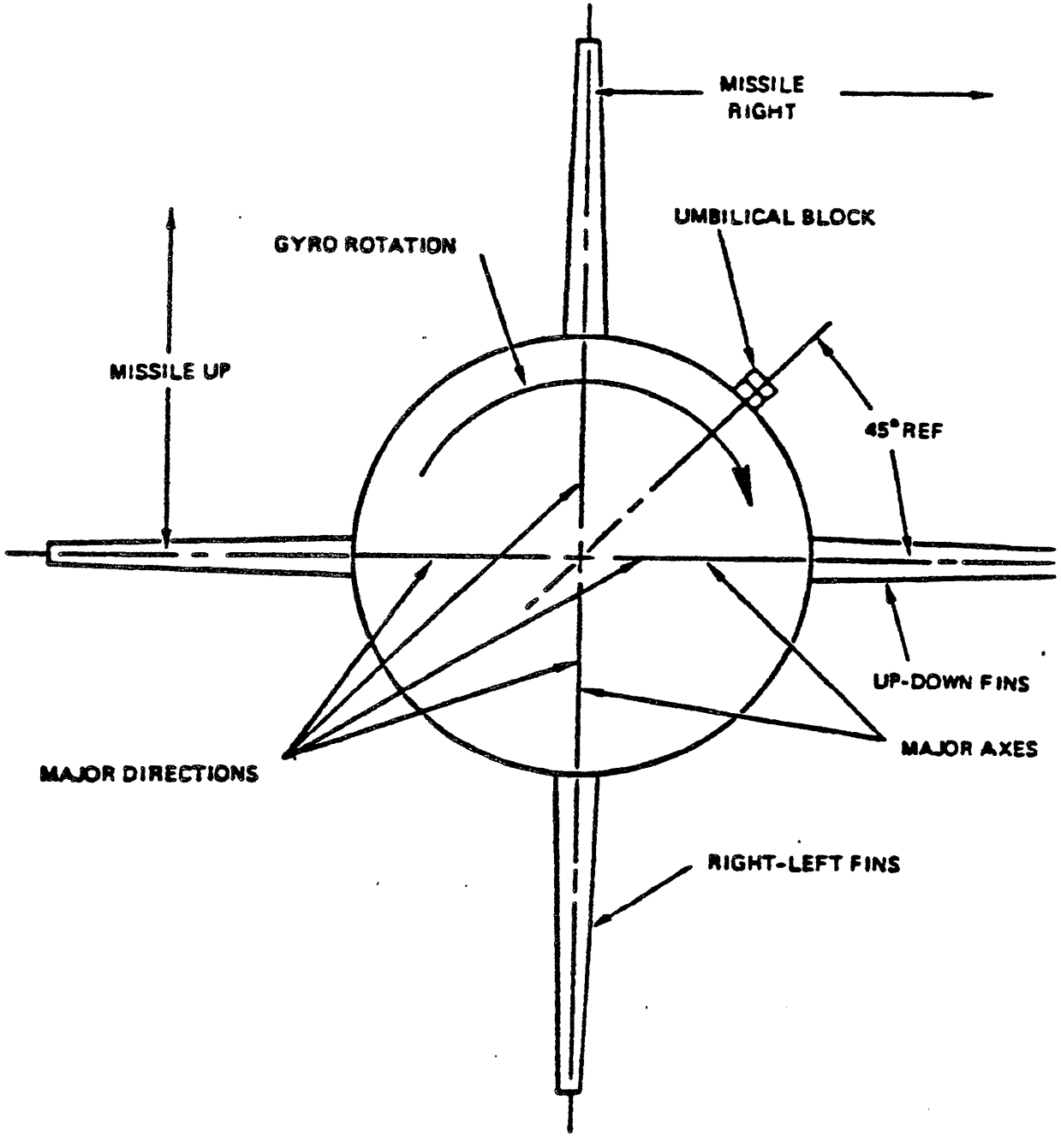


FIGURE 8. Guidance section orientation.

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4.3.3.2 Video output pulse amplitude. With the GCS tracking a target as specified in 4.3.3 except the power levels shall be threshold +10 db and attenuator enablement +3 dB, measure the video output pulse amplitude at A2W1J6 pin 8. The RF attenuator will be enabled when a non-zero voltage is detected at A2W1J6 pin 12. The threshold will be attained by increasing the power level until a steady (non-ramping) voltage is detected at A3A1WJP1 pin 12. Pulse amplitude shall meet the requirements of 3.2.1.6.

4.3.3.3 Target acquisition. With no target signal applied the GM shall attempt to acquire a target as observed by the local oscillator sweep voltage of 0 to 10 V at A3A1W1P1 pin 12. A target with signal conditions of 4.3.3 shall be turned on and the GCS shall acquire the target within 1.3 sec as observed by a steady voltage a A3A1W1F1 pin 12. After target acquisition, the uncage/cage command shall cause the GCS to re-scan and reacquire the target.

4.3.3.4 Target reacquisition. With the signal conditions of 4.3.3 applied, the GCS shall acquire a target as observed by a steady voltage at A3A1M1P1 pin 12. The target shall be turned off for 200 $\pm$ 50 ms. The GCS shall reacquire the target within 1.5 sec.

4.3.3.5 Field-of-view. The GCS in a caged mode shall be positioned such that a target is 15 deg off boresight while signal conditions of 4.3.3 are applied. Upon uncaging the GCS shall angle track the target as observed at A3A1W1P1 pin 14. This test shall be repeated at frequencies of A and G.

4.3.3.6 Lock-on sensitivity and dynamic range. With signal conditions of 4.3.3 applied except power level shall be 15 CD below design threshold, the GCS shall attempt to acquire a target at boresight ±3 deg while in the caged mode. The power level shall be increased until the GCS acquires the target as observed at A3A1W1P1 pin 12. The value of the power level shall be repeated for frequencies A, B, C, E, F, and G. This test shall also be repeated with the GCS in the uncaged mode except that after threshold is determined, the power level shall be increased until the RF attenuator is enabled as observed at A2W1J6 pin 12. The power level will then be decreased until the RF attenuator is again disabled. These power levels shall be recorded and meet the requirements of 3.2.1.11.

4.3.3.7 Gyro steady-state speed and slowdown time. With the gyro caged, measure the frequency of the gyro reference voltage (EL REF) pin 3 of A3AlwlP1. This frequency shall meet the requirements of 3.2.1.12. Remove the input power from the GCS. Measure the frequency 10 seconds after removal of the input power. This frequency shall meet the requirements of 3.2.1.12.

4.3.3.8 Loop gain/phasing.

4.3.3.8.1 Tracking loop gain and cross plate coupling tracking error. With the GCS tracking a target moving at a planar rate of 4 deg/sec in the uncaged state the tracking error shall be between 1.6 degrees and 0.67 degree for frequencies from A + 0.5 GHz to C, between 1.33 degrees and 0.67 degree for frequencies from C to E and between 1.33 degrees and 0.4 degree

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for frequencies E to G. This test shall be conducted at the frequencies of A + 0.5 GHZ through G - 0.5 GHz; polarizations of 0, 45, 90 and 135 degrees; PRF 1 for frequencies A + 0.5 GHz and G - 0.5 GRz and PRF 2 for frequencies B, C, D, E, and F; and pulse width 2. This test shall be conducted in both the azimuth and elevation planes. The in plane and at of plane values shall be determined from the average value of the 4 polarizations. The out of plane tracking error shall be less than 10 percent of the in plane tracking error. These requirements shall be met for the 10 of the 14 measurements.

4.3.3.8.2 Caging loop gain. with the GCS in the caged mode, rate the GCS at a planar angular race of 4 deg/sec. Measure the amount of gyro head position lag as the GCS is moving. The lag shall be less than 0.66 deg and greater than 0.4 deg. Make the measurement in the azimuth plane only (see Figure 10).

4.3.3.9 Pilot tone. With signal conditions of 4.3.3 applied, the seeker caged or uncaged and a target positioned on boresight, measure the pilot tone output at connector P1004 between pins 15 and 14 into a 20,000 ohm ±20 percent load. The signal shall meet the requirements of 3.2.1.14.

4.3.3.10 Servo system tests. Unless otherwise specified herein, the following tests specified in 4.3.3.10.1 through 4.3.3.10.3 shall be performed in each of the four quadrants with GCS on internal power, the magnetic amplifier enabled, G-bias disabled, and NAV gain disabled.

4.3.3.10.1 Servo torque output. Measure the servo torque output in four quadrants while tracking a target moving at a planar rate of 6 deg/sec. The torque shall meet the requirements of 3.2.1.15.

4.3.3.10.2 Servo torque balance. With the gyro caged and the G-bias disabled, measure the servo torque developed in each channel with the magnetic amplifier both enabled and disabled. Measure from zero torque to an average recorded value. The average value of servo torque shall not exceed the value specified in 3.2.1.16.

4.3.3.10 Servo output noise. With the signal conditions of 4.3.3 applied, track a stationary target positioned on an extension of the longitudinal axis of the GCS which is operating on internal power. Record the average servo torque developed in the left-right channel and the up-down channel. No more than five servo torque peaks in excess of 100 in-lb in any one direction, as measured from the average levels shall be observed during any 2 sec interval of a 6 sec sample.

4.3.3.11 System gain. System gain is considered as the change in the servo torque amplitude divided by the change in the rate of gyro procession. With signal conditions of 4.3.3 applied and G-Bias and NAV gain disabled, track a target positioned on an extension of the longitudinal axis of the GCS and moving at a planar rate of 1 deg/sec. The torque output shall be 270 ±70 in-lb in each of four quadrants.

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4.3.3.12 Average system gain for each channel. Compute average values of system gain for each channel: (right + left)/2 and (up + down)/2 using data from 4.3.3.11. The average value shall be 270 ±50 in-lb per deg/sec tracking rate.

4.3.3.13 Steady-state cross talk. With the Signal conditions of 4.3.3 applied and G-bias and NAV gain disabled, track a target moving at a planar rate of 6/deg/sec. Measure the servo torque developed in the up-down quadrant (when the tracking motion is in the right-left direction) and in the right-left quadrant (when the tracking motion is in the up-down direction). The p-p torque value of cross talk in the up-down direction when tracking the right-left quadrant shall be not greater than the requirements of 3.2.1.20. The p-p torque value of cross talk in the right-left direction when tracking in the up-down quadrant shall be not greater than the requirements of 3.2.1.20.

4.3.3.14 Servo disabled-enabled operation. With the signal conditions of 4.3.3 append G-bias and NAV gain disabled, track a target on an extension of the longitudinal axis of the GCS moving at a planar rate of 6 deg/sec. Measure servo torque output in each channel under the following conditions:

a. Changeover from external to internal power in such a way that enabling occurs when the tracking rate is 6 ±1 deg/sec. During operation on internal power and for at least 0.4 sec after removal of the magamp disabling voltage, the servo output torque in each channel shall met the requirements of 3.2.1.21.

b. Measure time after removal of the magamp disabling voltage for the servo torque to reach 65 percent of the steady-state value. This shall occur within the time limit specified in 3.2.1.21.

4.3.3.15 Enabling transient. Operate the GCS on internal power with the gyro tracking a target of power level 2C positioned on the GCS axis and a zero angular rate of the GCS. Alternately open and close the magamp enable-disable. The maximum amplitude of the enabling transient torque shall be as specified in 3.2.1.22.

4.3.3.16 G-bias nav/gain test. The G-bias test shall be performed to verify conformance to 3.2.1.23

a. Enable G-Bias, disable, NAV-gain and simulate power changeover. Measure the G-Bias vector torque for the three vectors as defined in the GCS mounting position phasing (see Figure 2). In each case, the fin deflection shall be in such a direction to cause an up maneuver with respect to the gravity vector and shall meet the requirements of 3.2.1.23.1.

b. Enable G-bias and NAV gain and simulate spewer changeover with the GCS in the umbilical up position (see Figure 2). The servo torque output shall conform the limits shown on Figure 3.

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4.3.3.17 Power changeover and umbilical disconnect. Operate the GCS on internal power with the magnetic amplifier disabled by applying +175 V between pin 31 and pin 35 of A3A1J1. Track a stationary target at power level 3B positioned on GCS axis. Simulate two umbilical disconnects by means of a discconnect relay (disconnect P1004 pins 1, 18, and 29). The test shall be made two times without loss of target track. These tests need not be performed in sequence but rather may be performed in the normal sequence of acceptance testing. Retention of target track can be verified by monitoring the APC voltage. Test results shall meet the requirement of 3.2.1.24.

4.3.3.18 Power video signal transients. Operate the GCS on internal power with magnetic amplifier enabled and no RF signal. Monitor the AFC voltage and signal boxcar. The AFC voltage should be a continuous sweep from 0 V to 10 V, repeating every 1.3 sec with no breaks. The boxcar should be a constant level with no breaks.

4.4 Temperature cycling and random vibration screenings. Prior to conducting the performance tests of 4.3.3, each CXS shall be subjected to the the tests Of 4.4.1 and 4.4.2.

4.4.1 Temperature cycling. Each GCS tested shall be subjected to not less than three temperature cycles. The temperature cycling shall be accomplished in the final GCS housing or in a tailoring housing provided that the housing is covered to minimize heat loss. The temperature range shall be -40 ± 5°C to +50 ±5°C. The temperature rate of change shall be not less than 10°C per minute. The temperature soak times shall be 2 hours at -40 ±5°C and 1.5 hours at +50 ±5°C. With the gyro caged, the GCS shall be operated continuously through each temperature cycle operating period. Servo air is not required. The GCS operating period for each cycle shall commence the last 30 minutes of the cold temperature soak and cease at the end of the high temperature soak as shown in Figure 9. GCS's that fail during temperature cycling shall be reworked and resubmitted to the temperature cycling screen at the beginning of the temperature cycle where the failure was detected. The last cycle shall be failure free. All failures shall be reported in accordance with the reliability program requirements of 3.2.2. Adjustment of tailoring resistors during temperature cycling shall not be considered a failure but must be reported to the procuring activity in writing (see 6.2.2). The following shall be monitored during the GCS on period:

a. Input current (see 3.2.1.1).

b. Applied voltage (see 3.2.1.1.1).

c. Gyro steady state speed (see 3.2.1.12).

d. AFC voltage (see 3.2.1.25).

e. Pilot tone (see 3.2.1.14).

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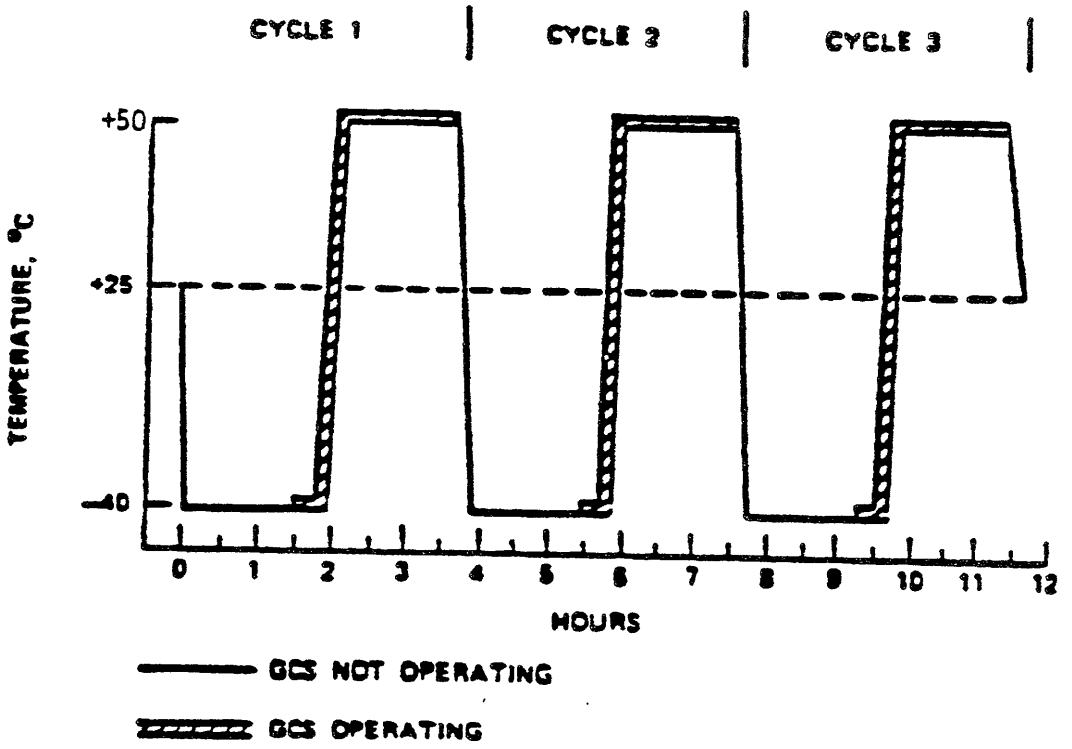


FIGURE 9. Temperature cycling screen.

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4.4.2 Random vibration screening. After completion of the temperature cycling of 4.4.1, the seeker section shall be removed and the assembled receiver and servo sections shall be subjected to random vibration screening. The receiver and servo shall-be mounted in a method approved by the procuring activity (see 6.2.2) and shall be subjected to random vibration along each of the three mutually perpendicular axes. The random vibration shall be for 10 minutes in each axis with the input shown on Figure 10. During the vibration test the receiver and servo shall be powered. The receivers and servos which fail to meet all performance requirements of 3.2.1 following vibration test, and require disassembly for rework, shall be subjected to an additional failure free temperature cycle only (see 4.4.1). The following must be monitored during the GCS on period:

a. Input current (see 3.2.1.1).

b. Applied voltage (see 3.2.1.1.1).

c. Pilot tone (see 3.2.1.14).

d. AFC signal (see 3.2.1.25).

e. Pilot tone (see 3.2.1.14).

4.5 Environmental tests. Immediately prior to performing the evnvironmental test specified herein, the GCS shall be subjected to the performance tests specified in 4.3.3 to establish a record of initial performance under standard conditions. The performance tests as specified in 4.3.3 shall also be performed at the conclusion of each environmental tests to determine system integrity and performance. These tests shall serve as initial record data for the next test providing the results are acceptable and within specified tolerances. These tests shall apply to first article only.

4.5.1 High temperature (nonoperating and operating). The GCS shall be tested in accordance with MIL-STD-810. Method 501.2, Procedures I and II. In Procedure 1, Step 4 shall consist of 48 hours of exposure to constant 71°C. Relative humidity shall not exceed 15 percent. It is not necessary to monitor any specific critical item during the 48 hours exposure. In Procedure II, the GCS shall be exposed to a temperature of 60°C with humidity not greater than 15 percent. Perform Steps 1, 7, 8, 9, 10, 15, 16 and 17 only. In Step 7, the GCS shall be allowed to stabilize at 60°C for 4 hours before conducing operational checkout. The GCS shall meet the requirements of 3.2.4.1.

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4.5.2 Low temperature (operating). The GCS shall be tested in accordance with a modification of MIL-STD-810, Method 502.2, Procedure I. Reduce the chamber temperature to -40°C and maintain this temperature until stabilization of the test item. After stabilization occurs, operate the test item and reduce the chamber temperature to -53°C within 5 minutes maximum. Maintain -53°C for 1 hour after chamber stabilization with the unit operating. The GCS shall meet the requirements of 3.2.4.2.

4.5.3 Temperature shock non-operating. The GCS shall be tested in accordance with MIL-STD-810, Method 503.2, Procedure 1, except the GCS shall be allowed to stabilize at each temperature for a period of 4 hours. The temperature extremes shall be +71°C and -40°C. The GCSM shall meet the requirements of 3.2.4.3.

4.5.4 Flight acceleration (operating). The GCS shall be separately tested in accordance with MIL-STD-410, Method 513.3, Procedure II. The acceleration levels shall be 25 gravity units (g) in the forward longitudinal direction, 10 g in the aft longitudinal direction, and 28 g in combined plane lateral axis (four directions). The GCS shall meet the requirements Of 3.2.4.4.

4.5.5 Transportation vibration (nonoperating). When specified in the contract or purchase order (see 6.2.1), the GCS, installed in a shipping container, shall be tested in accordance with MIL-STD-810, Method 514.3, Procedure I (modified). Step 6 shall be omitted. The spectrum shown on Figure 4 shall be applied for 3 hours in each of three orthogonal axes of the container shall meet the requirements of 3.2.4.5.

4.5.6 Captive flight vibration - fixed wiring aircraft (operating),

a. Maneuver. The GCS shall be tested in accordance with MIL-STD-810, Method 514.3, Procedure I. The vibration test level shall be as shown on Figure 5 applied for 12 minutes in each of the three axes. GCS shall meet the requirements of 3.2.4.6.

b. High-speed dash. The GCS shall be tested in accordance with MIL-STD-810, Method 514.3 Procedure I. The vibration shall be as shown on Figure 6 applied for 2.5 minutes in each axis. The GCS shall meet the requirements of 3.2.4.7.

4.5.7 Captive carry vibration - helicopter (operating). The GCS shall be tested MIL-STD-810, Method 514.2, Procedure I except that the vibration test levels shall be sinusoidal excitation at the frequencies and amplitudes shown in Table I. Each frequency shall be applied for 10 minutes per axis. The GCS shall meet the requirements of 3.2.4.8.

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4.5.8 Transit shock (nonoperating). When specified in the contract or purchase order (see 6.2.1), the GCS, installed in a shipping container, Ghan be tested in accordance with MIL-STD-810, Method 516.3, Procedure IV. The GCS shall meet the requirements of 3.2.4.9.

4.5.9 Ignition or tip-off shock (operating). The GM shall be tested in accordance with MIL-STD-810 Method 516.3. Procedure I except that the shock pulse inputs shall be 30 g and 8 msec, half sine in the forward direction of the longitudinal axis, and 35 g by 15 msec terminal sawtooth in both directions of the vertical and horizontal axes for a total of five shocks. The GCS shall meet the requirements of 3.2.4.10.

4.6 Failure reporting, analysis and corrective action. Unless otherwise specified in the contract or purchase order, the contractor shall implement and maintain a comprehensive system for identification, reporting, investigation, analysis, remedy, correction, and the prevention of recurrence of all problems and failures which occur throughout the production program (see 6.2.2).

4.6.1 Failure criteria. Failure categories are classified as relevant or nonrelevant.

4.6.1.1 Relevant failures. A relevant failure is any failure except chose classified as specified in 4.6.1.2.

4.6.1.2 Nonrelevant failures. Nonrelevant failures are defined as follows:

a. Any failure caused by test equipment.

b. Any failure that results from operational or procedural error in testing.

c. Any failure resulting from rework, repair, or investigatiOn Of failures during environmental or MTBF testing if detected prior to resubmission to environmental or MTBF testing.

d. Secondary failures when they are conformed to be induceed by failure of another part. However, at least one primary (relevant) failure must be charged when one secondary failure is claimed.

e. When a failure results from a design incompatibility which has been determined to exist and has been concurred by the procuring activity provided chat the previously required corrective action for the incompatibility does not apply to the lot in question.

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f. Repeated failures for which corrective action has been documented but, by direction of the procuring activity, the corrective action was not incorporated in the GCS lot where the repeated failures occur.

g. Failure of approved limited life parts when the failure occurs after the stipulated life period.

5. PACKAGING

5.1 Preservation, packaging and packing. For definitions of preservation, packaging, and packing levels see FED-STD-102. Unless otherwise specified in the contract or purchase orders (see 6.2.1), preservation and packaging shall be in existing AIM-9C shipping containers. The preservation and packaging shall include cleaning, dryings and preservation methods adequate to prevent deterioration; appropriate protective wrappings; package cushioning; and interior containers excluding shipping containers. Shipping container (Drawing 1516568), Mark 270 Mod 1, will be Government-furnished.

5.1.1 Level C. The prime contractor shall be responsible for the preservation packaging and packing of components received from subcontractors. Containers shall comply with Consolidated Freight Rules or other common carrier regulations applicable to the mode of transportation.

5.2 Packing. Packing shall be Level C.

5.2.1 Level C. Packing shall cover the exterior shipping container, the assembly of items or packages therein, necessary blocking, bracing, cushioning and weatherproofing.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2.1), all containers shall be marked in accordance with the applicable drawings, MIL-STD-129, and Code of Federal Regulations (CFR), 49 CFR 171-178.

5.3.1 Special marking. Units which have been submjected to the environmental tests of 4.5 shall be marked “DO NOT USE FOR TACTICAL PURPOSES". The container shall also be marked accordingly.

6. NOTES

6.1 Intended use. The GCS is intended for use as guidance-control element of the Sidearm air-to-ground guided missile.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Utilization of WS-6536 if different that 3.3.3.

c. The activity responsible for inspection (see 4.1).

d. Disposition of test samples (see 4.1.2).

e. Reliability demonstration if different from 4.1.2.3.

f. Environmental demonstration (see 4.1.2.4).

g. Transportation vibration if different from 4.5.5.

h. Transit shock (see 4.5.8).

i. Levels of preservation, packaging, and packing (5.1 and 5.2).

j. Marking if different than 5.3.

6.2.2 Data requirements. When this specification is used in an acquisition and data are to be delivered, the data requirements identified below shold be developed as specified by and approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.3.1	Reliability Program Plan	DI-R-7079	—
3.5	Elctromagnetic Compatiblity Program Plan	DI-R-7096	—
4.1.2.2	Reliability Demonstration Test Plan/Agenda/Report	DI-T-5204	—
4.1.2.3	Environmental Qualification Test Reports	DI-T-2072	—
4.4.1, 4.4.2	Failure Report	DI-R-21598	—

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DOD 5010.12-L AMSDL. Copies of DIDs required by the contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of two GCSs. The contracting officer should include specific instruction in acquisition documents regarding arrangement for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Safety precautions. The loading, assembly and handling of the explosives, subassemblies, and the finished items covered by this specification involve hazardous operation and therefore require suitable explosives safety precautions Standard safety precaution for explosive-loaded item are contained in DOD 4145.26M.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

6.6 Subject term (keyword) listing.

Air to ground
RF guidance
Passive
AIM-9C
Seeker
Acquisition(target)
Tracking(target)
Proportional navigation

Preparing activity
Navy-AS

(Project 1420-N285)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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